Interband Pairing Signatures in the Superfluid Response of Magic-Angle Twisted Bilayer Graphene

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The superfluid response of magic-angle twisted bilayer graphene is strongly affected by quantum geometry, but the role of pairing structure in this response remains less directly quantified. We construct an executable Bistritzer-MacDonald–Bogoliubov-de Gennes workflow for testing orbital-projected interband-pairing directions in a finite-band response calculation. The pairing family is defined in the layer-sublattice orbital basis and then projected into the normal-state band basis using a fixed time-reversal-sewn valley convention. For the working direction $M_1 = \tau_x\sigma_x$, the projected interband pairing weight increases smoothly with the tuning parameter η and reaches approximately 0.108 at $\eta = 1$ for the $n_{\mathrm{keep}} = 6$ truncation. Dense scans over chemical potential and pairing interpolation show that, at fixed Frobenius norm of the orbital pairing matrix, the normalized isotropic response changes by approximately $+0.56\%$ to $+11.86\%$ between $\eta = 0$ and $\eta = 1$ over $\mu = -5, \ldots, 5$ meV. The same scan at fixed raw gap amplitude is much weaker and can be negative. We therefore identify the signal as a normalization-conditioned mechanism signature rather than an unconditional enhancement of the absolute stiffness. Finally, we audit the absolute-unit baseline against a previous PRB-style superfluid-weight table and show that the historical table can be only partially reconstructed from a unified convention. This motivates using normalized response maps for the main mechanism claim and keeping absolute stiffness comparisons as a separate benchmark problem.

I. INTRODUCTION

Magic-angle twisted bilayer graphene (MATBG) provides a setting in which superconductivity, topology, flat bands, and quantum geometry meet in a single experimentally accessible platform [1–3]. In conventional superconductors the superfluid weight is tied to band dispersion and carrier density, but in flat or nearly flat bands a geometric contribution associated with the quantum metric can remain finite even when ordinary band velocity is small [4, 5]. This geometric contribution is especially relevant for MATBG, whose fragile topology and $C_{2z}\mathcal{T}$ symmetry protect nontrivial band geometry in the flat-band manifold [6–8].

Circuit-QED measurements have reported a superfluid stiffness in MATBG that is large compared with a simple BCS estimate [9]. More recent transport and microwave measurements have directly extracted the superfluid stiffness and found values above conventional Fermi-liquid expectations, while also pointing to nontrivial low-temperature gap structure [10]. Several theoretical works have connected this anomaly to quantum geometry, remote-band effects, and beyond-BCS corrections [11–13]. Yet a practical question remains: if a mean-field pairing ansatz contains interband structure after projection into the MATBG band basis, does that structure leave a robust and separable signature in the computed superfluid response?

This paper addresses that question with an explicitly executable finite-band workflow. Rather than beginning from a band-basis off-diagonal gap chosen by hand, we define a pairing matrix in the layer-sublattice orbital basis and project it into the normal-state band basis. This construction makes the pairing direction covariant under band-basis gauge changes and separates two issues that are often entangled: the definition of interband pairing and the response convention used to report stiffness. The main output is a normalized mechanism map,

$$\delta D_{\mathrm{iso}}(\mu, \eta) = \frac{D_{\mathrm{iso}}(\mu, \eta)}{D_{\mathrm{iso}}(\mu, 0)} - 1, \tag{1}$$

computed for two gap-normalization controls.

The central result is deliberately conservative. At fixed Frobenius norm of the orbital pairing direction, increasing the interband-pairing component produces a reproducible change in the isotropic response across chemical potential. At fixed raw gap amplitude, the effect is much smaller. Thus the present calculation supports a normalization-conditioned mechanism signature, not a universal statement that interband pairing always enhances the physical superfluid stiffness. We also perform a self-audit of the absolute units and a previous PRB-style benchmark table; that audit shows why absolute stiffness tables should be regenerated from a single declared convention before being used for final experimental comparison.

II. MODEL AND PAIRING CONSTRUCTION

A. Bistritzer-MacDonald model

We use a single-valley Bistritzer-MacDonald continuum Hamiltonian with momenta in Å^{-1} and energies in

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meV [14]. The basis for each moire reciprocal vector $\mathbf{G}$ is

$$(tA, tB, bA, bB), \tag{2}$$

where t, b label top and bottom layers and A, B label sublattices. Unless otherwise stated, the numerical scans use

$$\begin{aligned} \theta &= 1.05^\circ, & v_F &= 2.135 \text{ eV Å}, \\ w_0 &= 87.2 \text{ meV}, & w_1 &= 109.0 \text{ meV}, \end{aligned} \tag{3}$$

with a square reciprocal cutoff $N_{\text{shell}} = 3$, giving a 196×196 normal-state Hamiltonian per valley and spin. The moire unit-cell area and reciprocal scale used in the unit audit are

$$A_M = 1.5606 \times 10^4 \text{ Å}^2, \quad G_M = 0.054047 \text{ Å}^{-1}. \tag{4}$$

B. Orbital-defined pairing family

The pairing matrix is first specified in the orbital basis. We write

$$\Delta_{\text{orb}}(\eta) = \Delta_0 \mathcal{N}_\eta \left(M_0 + \eta M_1 \right), \tag{5}$$

with

$$M_0 = \tau_0\sigma_0, \qquad M_1 = \tau_x\sigma_x. \tag{6}$$

Here τ_i act in layer space and σ_i act in sublattice space. The parameter η interpolates between the reference orbital-singlet direction and the interlayer-sublattice-mixed direction. The normalization $\mathcal{N}_\eta$ is treated as a control variable. We use two conventions:

$$\text{fixed Frobenius:} \qquad \|M_0 + \eta M_1\|_F = \|M_0\|_F, \tag{7}$$

$$\text{fixed } \Delta_0: \qquad \mathcal{N}_\eta = 1. \tag{8}$$

At each momentum, the orbital pairing is projected into the retained band subspace as

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k}) \Delta_{\text{orb}} U_-^*(-\mathbf{k}). \tag{9}$$

The working valley convention is a time-reversal-sewn proxy, $U_-(-\mathbf{k}) = U_+^*(\mathbf{k})$. Thus the production scans use $\Delta_{\text{band}} = U_+^\dagger \Delta_{\text{orb}} U_+$ after the valley basis is sewn. A raw independently diagonalized valley-minus basis is not used for W_{inter} because it is not automatically in this sewing convention. This is used as a controlled diagnostic convention rather than a claim that all valley-basis subtleties are solved. The projected interband-pairing weight is

$$W_{\text{inter}} = \frac{\sum_{n \neq m} |\Delta_{nm}|^2}{\sum_{n,m} |\Delta_{nm}|^2}. \tag{10}$$

III. FINITE-BAND RESPONSE

For each retained band subspace we construct a finite-band BdG Hamiltonian,

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) - \mu & \Delta_{\text{band}}(\mathbf{k}) \\ \Delta_{\text{band}}^\dagger(\mathbf{k}) & -[\varepsilon(\mathbf{k}) - \mu] \end{pmatrix}. \tag{11}$$

The static current-current response is evaluated at zero temperature as

$$\Lambda_{\alpha\beta} = \sum_{a \neq b} \frac{\langle a|J_\alpha|b\rangle\langle b|J_\beta|a\rangle(f_a - f_b)}{E_b - E_a}. \tag{12}$$

The values reported by the executable response engine are finite-band raw diagnostics. They are internally consistent response outputs, but they are not used here as final experimental stiffness values.

The band-basis velocity is decomposed into diagonal and off-diagonal parts,

$$v_\alpha^{\text{band}} = v_\alpha^{\text{intra}} + v_\alpha^{\text{inter}}. \tag{13}$$

This yields conventional, geometric, and cross response channels. The isotropic response used in the mechanism scans is

$$D_{\text{iso}} = \frac{1}{2}(D_{xx} + D_{yy}). \tag{14}$$

The numerical checks track BdG Hermiticity, particle-hole symmetry, and exact closure of the channel decomposition.

IV. PAIRING GATES AND RESPONSE MAPS

A. Pairing and velocity gates

The first algebraic pairing gate verifies that an orbital-defined pairing family transforms covariantly under band-basis gauge changes. The BM projection gate then selects $M_1 = \tau_x\sigma_x$ as a useful working direction because it produces a finite projected interband weight in the time-reversal-sewn valley convention while keeping $W_{\text{inter}} = 0$ for $\eta = 0$. For $n_{\text{keep}} = 6$, $nk = 7$, and $\eta = 1$, the dense scan gives

$$W_{\text{inter}} \simeq 0.108. \tag{15}$$

The velocity gate confirms that projected velocity matrices are Hermitian to machine precision and that the intra/inter velocity split closes numerically.

B. Dense chemical-potential and pairing scan

Table I summarizes the dense scan at $nk = 7$ and $n_{\text{keep}} = 6$. The first column gives the retained-band filling proxy ν_{proxy}; this is a diagnostic occupancy proxy and is not used as a calibrated experimental filling.

TABLE I. Dense μ scan comparing $\eta = 1$ to $\eta = 0$. The fixed-Frobenius column reports $D_{\mathrm{iso}}(\eta = 1)/D_{\mathrm{iso}}(\eta = 0) - 1$. The fixed-$\Delta_0$ column reports the same quantity without rescaling the orbital pairing matrix.

μ (meV)	ν_{proxy}	fixed Frobenius	fixed Δ_0
-5	-1.483	0.0056	-0.0510
-4	-1.333	0.0891	0.0291
-3	-1.061	0.0785	-0.0071
-2	-0.857	0.0537	-0.0110
-1	-0.653	0.0802	0.0093
0	-0.381	0.0817	0.0112
1	-0.054	0.0914	0.0044
2	0.122	0.0949	0.0119
3	0.395	0.0653	0.0004
4	0.667	0.0662	-0.0005
5	0.857	0.1186	0.0312

TABLE II. $nk = 9$, $n_{\mathrm{keep}} = 6$ validation of the $\eta = 1$ response.

μ (meV)	fixed Frobenius	fixed Δ_0
-4	0.0604	-0.0068
0	0.0863	0.0103
2	0.0643	-0.0029
4	0.0605	-0.0017

Figure 1 shows the fixed-Frobenius dense response map. The enhancement is positive over most of the scanned chemical-potential range, with the largest endpoint value in the present scan occurring near $\mu = 5$ meV. The middle panel shows that the geometric fraction remains a useful diagnostic of how the response is redistributed between current sectors.

Figure 2 shows the same scan at fixed raw Δ_0. The response is much weaker and can be negative. This contrast is the main reason that the conclusion is stated as normalization-conditioned.

C. $nk = 9$ validation

The key-point validation in Table II repeats selected values at $nk = 9$. The fixed-Frobenius signal remains positive and of comparable size, while the fixed-Δ_0 response stays close to zero. Additional selected $nk = 11$ and $nk = 13$ key-point checks, reported in the Supplemental Material, preserve the positive fixed-Frobenius response at the same chemical potentials.

V. SELF-AUDIT OF ABSOLUTE UNITS

The raw response engine outputs values in finite-band diagnostic units. A conservative conversion maps raw meV Å^2 to eV Å^2 per flavor by dividing by 1000. Any spin/valley degeneracy factor is left outside this conversion:

$$D_{\mathrm{per\ flavor}}[\mathrm{eV\,\AA}^2] = D_{\mathrm{raw}}[\mathrm{meV\,\AA}^2]/1000. \tag{16}$$

The moire area and reciprocal scale agree with the standard MATBG geometry, but the historical PRB-style benchmark table is not reproduced by a single global prefactor or by the present production convention.

Table III summarizes the reconstruction audit for uniform s-wave pairing at $\mu = 0$, $\Delta_0 = 1$ meV, and $nk = 14$. The best simple unified reconstruction is

$$D_{\mathrm{conv}} = 2D_{\mathrm{intra},\tau_z}, \qquad D_{\mathrm{geom}} = D_{\mathrm{inter},\tau_z}. \tag{17}$$

It nearly reconstructs the extended-band endpoint but not the flat-band endpoint. A separate flat-band audit shows that the old $n_{\mathrm{keep}} = 2$ total and conventional values can be approximated with a Gamma-centered mesh and a full curvature-like conventional correction, but that correction does not work cleanly at $n_{\mathrm{keep}} = 6$.

This self-audit changes the interpretation of absolute stiffness values. The historical table is useful as a benchmark, but it should not be used to set the convention for the present interband-pairing mechanism scans. Future absolute tables should be regenerated from one declared response convention, with degeneracy, Brillouin-zone normalization, and any diamagnetic or curvature term stated explicitly.

VI. DISCUSSION

The main physical message is that an interband-pairing direction can leave a detectable imprint on the normalized finite-band response of MATBG. The imprint is reproducible across chemical potential and survives a modest increase of the momentum grid, but it depends strongly on how the orbital gap direction is normalized. This is both a result and a warning: interband pairing

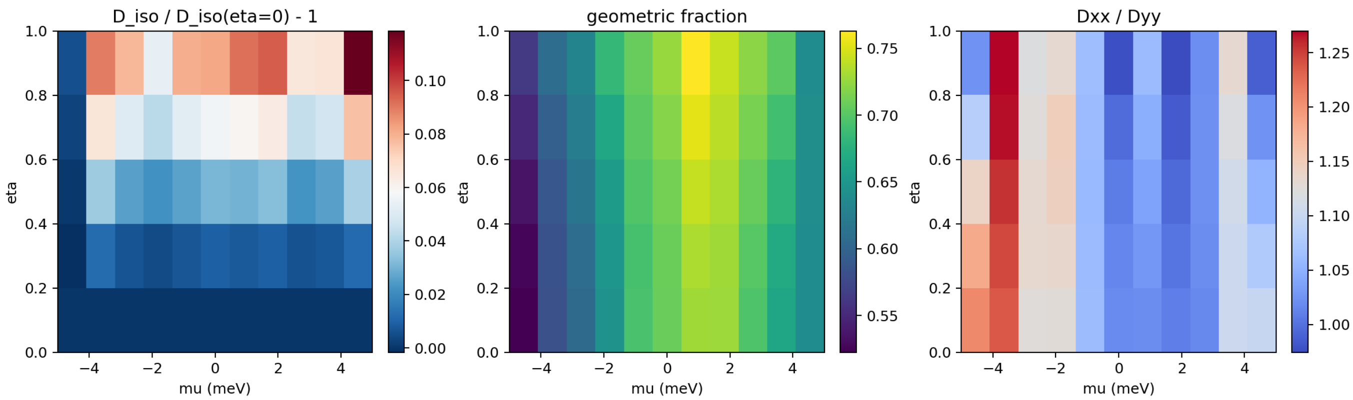

FIG. 1. Dense μ-η response map in the fixed-Frobenius normalization. Left: normalized isotropic response relative to $\eta = 0$ at fixed μ. Middle: geometric fraction. Right: response anisotropy D_{xx}/D_{yy}.

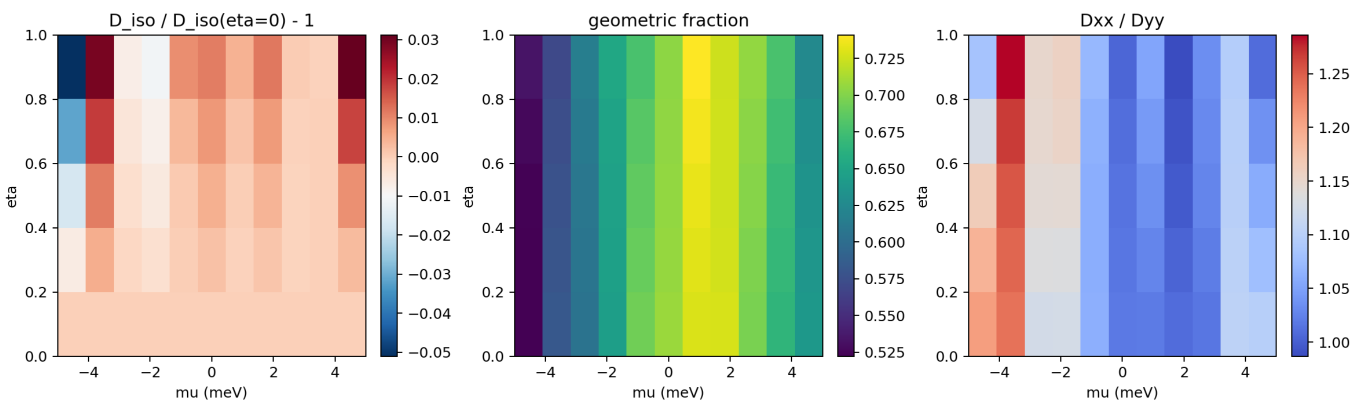

FIG. 2. Dense μ-η response map in the fixed-Δ_0 normalization. Compared with Fig. 1, the normalized response is strongly suppressed and changes sign in parts of the scan.

is not a scalar knob. Changing the orbital structure can also change the total norm of the projected gap, and the resulting response should not be interpreted without specifying the normalization convention.

The self-audit also clarifies the status of absolute stiffness claims. The continuum BM model has subtleties associated with UV regularization, current convention, and band truncation. Those subtleties are manageable, but the audit shows that old absolute tables can mix conventions in ways that are not captured by one simple prefactor. For this reason the most robust near-term observable in the present work is a normalized response map, not an absolute stiffness prediction. Likewise, ν_{proxy} in the tables is a retained-band occupancy proxy defined for scan organization, not a calibrated carrier density. The present figures should therefore be read as functions of chemical potential and retained-band occupancy, not as a quantitative map onto an experimental superconducting dome.

Several limitations remain. First, the valley sewing convention used here is a controlled time-reversal proxy; a fully independent two-valley implementation would be required for valley-asymmetric perturbations or final absolute production calculations. Second, the absolute response convention is not yet final for experimental stiffness comparison. Third, the filling variable is currently a retained-band occupancy proxy. Fourth, no self-consistent gap equation is solved. Fifth, a UV-complete lattice calculation would be required for final quantitative comparison with experiment. Finally, recent finite-momentum Kekulé or pair-density-wave scenarios connect tunneling spectra, gapless quasiparticles, and geometric stiffness in twisted graphene [15]. Those states are complementary to the present uniform-pairing diagnostic and will require a separate finite-momentum BdG extension.

VII. CONCLUSIONS

We have built an executable MATBG response pipeline for testing orbital-projected interband-pairing signatures. The working pairing direction $M_1 = \tau_x\sigma_x$ produces a finite projected interband-pairing weight and a reproducible normalized response in the fixed-Frobenius scan. The same mechanism is strongly suppressed when

TABLE III. Historical PRB benchmark reconstruction. Values are eV Å^2 per flavor. The reconstruction route is kept separate from the production interband-pairing scans.

case	n_{keep}	D_{total}	target	D_{conv}	target	D_{geom}
current τ_z/τ_0	2	29.34	67.50	20.95	53.00	8.40
double-conv all-τ_z	2	54.40	67.50	41.89	53.00	12.50
flat endpoint audit	2	66.18	67.50	53.83	53.00	12.36
double-conv all-τ_z	6	126.66	129.30	51.45	54.00	75.22

the raw gap amplitude is held fixed. Thus the calculation supports a precise and conservative conclusion: interband pairing produces a normalization-conditioned mechanism signature in the finite-band MATBG superfluid response.

The accompanying benchmark audit shows that absolute stiffness tables should be regenerated from a single modern convention before being used as final manuscript or experimental-comparison numbers. The normalized response maps reported here provide the more robust foundation for the next stage of the PRB manuscript.

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